

Task 3: API Security Risk Analysis Report

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1. Objective

To analyze API security risks, specifically focusing on authentication, authorization, and data exposure vulnerabilities, and to provide remediation suggestions.

2. API Security Analysis

During the analysis of sample API endpoints, the following security risks were identified:

- **Broken Authentication & Authorization:** Inadequate validation of authentication tokens and API keys allows unauthorized users to access restricted resources.
- **Data Exposure:** Sensitive data is transmitted in plain text or is accessible through insecure endpoints due to a lack of proper data masking.
- **Missing Rate-Limiting:** The absence of rate-limiting mechanisms leaves API endpoints vulnerable to brute-force attacks and automated abuse.
- **Input Validation Gaps:** Insufficient validation of input data allows for malicious payload injection, potentially leading to injection attacks.

3. Remediation Recommendations

To secure API endpoints, the following best practices should be implemented:

- **Strengthen Authentication:** Enforce robust OAuth/OpenID Connect flows and ensure secure, short-lived token management.
- **Implement Authorization:** Apply the principle of least privilege by ensuring every API request is validated against the user's permissions.
- **Enforce Rate-Limiting:** Implement strict rate-limiting policies to prevent abuse and denial-of-service (DoS) attempts.
- **Sanitize Input:** Perform rigorous server-side input validation and encoding to mitigate injection risks.